Nicolas Zucchet

PhD Student · ETH Zürich

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Education

Postdoctoral scholar — *Stanford, USA*
Stanford — *Incoming*
- SNSF Postdoc mobility fellow in Prof. Dr. Scott Lindermann's lab.

PhD student — *Zürich, Switzerland*
ETH Zürich — *09/2021 - 09/2025*
- Doctoral student in artifical intelligence and computational neuroscience in the lab of Prof. Dr. Angelika Steger, jointly supervised by Dr. João Sacramento.

Computer science MSc — *Zürich, Switzerland*
ETH Zürich — *09/2019 - 08/2021*
- Theoretical foundations of artificial intelligence (optimization, neuroscience, machine learning).
- Master thesis: "Equilibrium propagation for bilevel optimization", supervised by Dr. João Sacramento.

Ingénieur polytechnicien (MSc) — *Palaiseau, France*
École polytechnique — *09/2016 - 08/2020*
- Multidisciplinary education (applied mathematics, computer science, economics), specialization in deep learning and computer vision during the last two semesters.

Classe préparatoire aux grandes écoles (MPI*) — *Versailles, France*
Lycée Hoche — *09/2014 - 08/2016*
- Intensive trainining for competitive entrance exams to french Grandes Écoles (fundamental mathematics, computer science, physics).

Professional experience

Student researcher — *London, United Kingdom*
Google DeepMind — *09/2024-02/2025*
- Research internship with Dr. Soham De, studying the learning dynamics of language models.

Visiting student — *Bern, Switzerland*
University of Bern — *05/2021 - 08/2021*
- Probabilistic methods for continual learning, hosted by Prof. Dr. Jean-Pascal Pfister.

Research internship — *Paris, France*
Prophesee — *05/2019 - 08/2019*
- Deep learning algorithms for event-based cameras in autonomous cars.

Software developer intern — *Sophia-Antipolis, France*
Amadeus — *06/2018 - 08/2018*
- Design and development of continuous integration tools.

Publications

Preprints

[3] Teaching signal synchronization in deep neural networks with prospective neurons, **N. Zucchet***, Q. Feng*, A. Laborieux, F. Zenke, W. Senn†, J. Sacramento†.

[2] Uncovering mesa-optimization algorithms in Transformers, J. von Oswald*, E. Niklasson*, M. Schlegel*, S. Kobayashi, **N. Zucchet**, N. Scherrer, N. Miller, M. Sandler, B. Agüera y Arcas, M. Vladymyrov, R. Pascanu and J. Sacramento.

[1] Gated RNNs discover attention, **N. Zucchet***, S. Kobayashi*, Y. Akram*, J. von Oswald, M. Larcher, A. Steger†, J. Sacramento†.

Conference and journal papers

[8] The emergence of sparse attention: impact of data distribution and benefits of repetition, **N. Zucchet**, F. D'Angelo, A. Lampinen, S. Chan. 39th Conference on Neural Information Processing Systems (NeurIPS), 2025.

[7] How do language models learn facts? Dynamics, curricula and hallucinations, **N. Zucchet**, J. Bornschein, S. Chan, A. Lampinen, R. Pascanu, S. De. 2nd Conference on Language Modeling.

[6] Recurrent neural networks: vanishing and exploding gradients are not the end of the story, **N. Zucchet**, A. Orvieto. 38th Conference on Neural Information Processing Systems (NeurIPS), 2024.

[5] Online learning of long-range dependencies. **N. Zucchet***, R. Meier*, S. Schug*, A. Mujika and J. Sacramento. 37th Conference on Neural Information Processing Systems (NeurIPS), 2023.

[4] The least-control principle for local learning at equilibrium. A. Meulemans*, **N. Zucchet***, S. Kobayashi*, J. von Oswald and J. Sacramento. 36th Conference on Neural Information Processing Systems (NeurIPS), 2022.

[3] A contrastive rule for meta-learning. **N. Zucchet***, S. Schug*, J. von Oswald*, D. Zhao and J. Sacramento. 36th Conference on Neural Information Processing Systems (NeurIPS), 2022.

[2] Beyond backpropagation: Bilevel optimization through implicit differentiation and equilibrium propagation. **N. Zucchet** and J. Sacramento. Neural Computation 34 (12), 2022.

[1] Learning where to learn: Gradient sparsity in meta and continual learning. J. von Oswald*, D. Zhao*, S. Kobayashi, S. Schug, M. Caccia, **N. Zucchet** and J. Sacramento. 35th Conference on Neural Information Processing Systems (NeurIPS), 2021.

Workshop papers

[1] Random initialisations performing above chance and how to find them. F. Benzing, S. Schug, R. Meier, J. von Oswald, Y. Akram, **N. Zucchet**, L. Aitchison†, A. Steger†. OPT2022: 14th Annual Workshop on Optimization for Machine Learning (NeurIPS), 2022.

Open source code bases

[1] Minimal LRU (61 stars). N. Zucchet, R. Meier, S. Schug.

[2] Online learning of long-range-dependencies (20 stars). N. Zucchet, R. Meier, S. Schug, A. Mujika, J. Sacramento.

Talks

Dec. 2025	**The emergence of sparse attention: impact of data distribution and benefits of repetition**, NeurIPS.
Nov. 2025	**How does data shape learning? A case study of factual recall and the role of data diversity**, MPI Tübingen.
Oct. 2025	**How do language models learn facts? Dynamics, curricula, and hallucinations**, COLM.
Dec. 2024	**Learning in the brain: what can we learn from deep state-space models?**, Linderman's lab, Stanford.
Jan. 2024	**Online learning of long-range dependencies**, Swiss Computational Neuroscience Retreat.
Feb. 2023	**The least-control principle for local learning at equilibrium**, Swiss Computational Neuroscience Retreat.
Nov. 2022	**The least-control principle for local learning at equilibrium**, Jean-Rémi King's Brain & AI group at Meta AI.
July 2022	**Biologically plausible bilevel optimization**, Rafal Bogacz's group at Oxford.
June 2022	**Bilevel optimization in neural networks**, MLSS^N 2022 lecture with J. Sacramento, A. Meulemans and S. Schug.
Feb. 2022	**A contrastive rule for meta-learning**, Swiss Computational Neuroscience Retreat.

Teaching experience

2022-25	**Algorithms lab**, Teaching assistant (Master's course).
2022-24	**Algorithms and probability**, Teaching assistant (Bachelor's course).
2021	**Randomized algorithms**, Teaching assistant (Master's course).
2016-17	**High-school mathematics**, Full-time tutoring of high school students from deprived neighborhoods.

Mentoring

2023 **Kevin Lopez**, Master thesis, ETH Zürich (co-supervised with J. Sacramento).
2023-24 **Yanick Schimpf**, Bachelor thesis, ETH Zürich.
2022-23 **Qianqian Feng**, Research project, ETH Zürich (co-supervised with J. Sacramento).
2022 **Anja Surina**, Master thesis, ETH Zürich (co-supervised with J. Sacramento and S. Schug).

Community service

2023-26 **ICLR**, Reviewer.
2023-25 **ICML**, Reviewer.
2023-25 **NeurIPS**, Reviewer (Best reviewer award in 2024).
2025 **TMLR**, Reviewer.

Awards

2025 **Postdoc mobility fellowship**, Swiss National Science Foundation (134’000 CHF).
2022 **Scholar award**, NeurIPS.